

CareCart: A Solution for College Students & Grocery Shopping Struggles

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Abstract

For many college students, grocery shopping is a stressful experience. With only a limited amount to spend each week, students also face the pressures of university life, and the challenges of managing health needs, from medical conditions to food intolerances. In the grocery store, this often leaves them feeling overwhelmed and unsure of what to buy that will fit both their budget and their diet. This stress not only affects their academic focus and mental well-being, but also contributes to poor eating habits and long term health outcomes.

Our project, CareCart, was created to address these everyday struggles. Care-cart is a student centered platform that suggests affordable, diet friendly alternatives while also offering a community space for recipe sharing and peer support. Using human-computer interaction methods and design thinking, we interviewed students, identified pain points, and prototyped starting from sketches, to a high-fidelity platform. With usability heuristics in mind, we designed CareCart to emphasize clarity and ease of use. More than just a shopping tool, it is designed to make grocery shopping simpler, healthier, and more supportive for students.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *User studies*.

Keywords

human-computer interaction, user-centered design, prototyping, usability testing, nutrition, student wellness

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1 Introduction

Grocery shopping is a routine part of our life, but for many college students it can quickly become a very stressful experience. Often,

with limited budgets, students find themselves standing in grocery store aisles making tough choices between what foods they want, need, and what they can actually afford. However, this challenge goes way beyond just the money aspect. For example, take a student managing PCOS who needs to watch their sugar intake, or someone who just figured out they're gluten intolerant. Now add midterms, a work schedule, and maybe trying to maintain some kind of social life. What should be a straightforward grocery run becomes an exhausting mental exercise that includes reading every nutrition label, doing mental math, and wondering why eating well has to be so complicated.

This pattern of falling back on convenient, processed foods builds up over time, leaving students feeling guilty, stressed, and frustrated about their inability to develop healthy habits. Existing apps also do not help entirely, as MyFitnessPal will tell you to eat more salmon and avocados without considering you have a small budget for the rest of the week, while Yuka points out that your granola bar is loaded with sugar but doesn't help you find affordable alternatives. Grocery list apps keep you organized but assume you already know what healthy and affordable food looks like.

This is why we created CareCart. We built a platform that actually understands what it's like to be a student trying to eat well on a tight budget. CareCart helps students find food alternatives that work for both their wallet and dietary needs, while connecting them with peers facing similar challenges. We used Human-Computer Interaction methods and design thinking, starting by talking to students about their real grocery shopping experiences. We heard stories about feeling overwhelmed in aisles, choosing between medication friendly foods and rent money, and the guilt of another week of unhealthy but affordable meals. After analyzing these conversations, we identified patterns and created personas representing different student shoppers. We built and tested prototypes through multiple rounds of feedback, using usability principles to ensure the platform was simple and straightforward, even when students felt stressed. CareCart addresses both financial reality and health considerations that students face, offering practical solutions while demonstrating how powerful design becomes when you truly understand what users are going through.

2 Background

College students today face demanding schedules, financial constraints, and inconsistent access to healthy food options- often at the expense of their physical and mental well-being. With limited time and resources, many rely on fast or processed foods, skip meals, or eat irregularly - all habits that contribute to digestive issues (such as

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Gastroesophageal Reflux Disease (GERD), constipation, and abdominal pain (Western Michigan University), as well as mental-health challenges (Kathleen Harris, MD). For students managing specific health conditions, food intolerances, and cultural/dietary restrictions, these challenges are even more pronounced. Despite growing awareness of nutrition's impact on wellness, students lack real-time, personalized tools that fit their daily needs, presenting an opportunity for a human-centered solution that supports healthier eating habits while fostering community and shared learning. CareCart aims to address this gap by helping students shop, make smarter & safer food choices, and eat with confidence by tailoring guidance to their unique health profiles and lifestyle constraints while also building a supportive network of peers.

3 Design method

To solve this problem, we followed the human-centered design process. First, we empathized with our users by interviewing three students: Sophie, Alex and Linda and asked each of them ten open-ended questions. We wrote down their open-ended responses and personal stories. Our interview sheet was carefully designed to follow a logical flow, moving from broad questions about meal planning and grocery shopping to targeted questions about dietary restrictions, health concerns, and their emotional experience with food. Finally, we explored technology use and expectations, asking about prior experience with nutrition apps and what features they would find helpful. Then we used thematic analysis from three paragraphs of quotes to identify pain points of users. More specifically, we found six pain points of potential users regarding their eating experience: time and energy constraints, special physical conditions, irregular meals, duplicate food choices, price concerns, and unclear chopping guidance. There are many intersections for time and energy concerns, and worries for unclear shopping guidance. Three participants all “do not have much time to cook”, which aligns with the research of current college students’ daily workout and stress. Additionally, we found that they all felt confused for “food labels”, and “overwhelming current food apps”, and considered shopping as “stressful”. The three main struggles of the interviewed three college students: lack of time leading to skipped meals, confusing food labels and hidden ingredients, and overwhelming nutrition apps.

We then defined our problem more clearly using persona and user journey maps. From our research, we gathered the information from the interviews and then curated 2 key personas - Linda and Alex. We created a persona for Linda as a 22 years-old pre-med student studying at UC Davis, representing the participants who complained “I’m always busy, but I still want quick and healthy meals that keep me going”. Her demanding academic schedule made her often have little time for cooking or grocery shopping while she knows the importance of nutrition. Linda pointed out frustrations for skipped meals and being overwhelmed when using complicated nutrition apps. These users expected to make grocery shopping smarter and faster, swapping for healthier food. Alex represented the participants who have special physical conditions like lactose intolerance and limited budget. He struggles with hidden ingredients due to dietary restrictions. These users expect an app that filters food by restrictions, suggests dairy-free alternatives, and

saves money with student-friendly recipes. We also mapped user journeys to better understand their frustrations: a female college student is frustrated by the meal plan for today during her busy schedule, and she goes to the supermarket quickly and doesn’t want to waste much time. When she arrives at the grocery store, she is confused by food labels and cannot figure out what nutrition each food contains. Finally, she ends up only taking chips and fries and continues to study while eating chips and fries.

Next we moved to ideation. Using divergent thinking, we brainstormed over 20 different ideas, including barcode scanning, affordable swaps, recipe sharing, and personalized meal planning, then applied convergent thinking and affinity diagramming to narrow them down to five key aspects: personalization & safety, affordability & budget, health guidance, meal planning & recipes, and behavior change & engagement. Specifically, for personalization, we considered allowing users to personalize their own health profile for dietary restrictions and allergies. Then we concluded these ideas into four core principles: personalization, affordability, simplicity and community. This sparked creativity but also ensured we stayed focused on the most impactful solutions. Then we worked on our prototypes.

Finally, we tested extensively with real users. We got three responses about their user experience from three real users Daniel, Priya, and Marcus. Through thematic analysis, we found that users enjoyed the community sharing feature, clean layout, fast scanner and traffic light system. But they still found our initial meal planning feature overwhelming, frustrated with expensive alternatives, and annoying reminders. Users prefer optional reminders over mandatory reminders, cheaper options and adaptive meal suggestions. This guided our refinements.

4 Understanding the users

In the early stages of our project, we focused on needfinding activities to understand college students’ struggles with healthy eating. Our target users were UC Davis students who face busy schedules, limited budgets, and inconsistent access to healthy food. Within this group, we paid special attention to students managing health conditions, dietary restrictions, and cultural food needs, since their challenges are often more pronounced. To collect data, we conducted interviews with three students and designed our interview sheet to follow a logical structure. We began with open-ended questions about meal planning and grocery shopping, barriers such as dietary restrictions and affordability, and concluded with technology-related questions about their experience with nutrition apps. Interviews allowed us to explore participants’ daily eating habits, practical barriers, and emotional feedback about food. We then recorded participants’ answers and analyzed them using thematic analysis and affinity diagramming. This helped us identify repeated frustrations such as time pressure, irregular meals, confusing labels, hidden ingredients, affordability concerns, and frustration with current apps. Through this process, we learned that students want simple, personalized, and accessible tools that save time, reduce stress, and provide trustworthy guidance. They also expressed a desire for community-driven features, such as peer recipe sharing, to make healthy eating more engaging. These findings directly informed our personas, problem statement, and later design decisions.

5 Conceptual Models in our Design

Carecart’s conceptual design is built around 3 key HCI design elements: interface metaphors, interactions, and different interfaces that all work together to simplify grocery shopping for college students.

The first element, interface metaphors, is the idea of using familiar real-world concepts to help users understand how to use a system. CareCart applies this element by implementing the idea of a “personal assistant” as a metaphor for a doctor or peer that understands both your health needs and budget constraints. This connection draws on the comfortable experience of shopping with someone who is efficient, understanding of your needs, and has the ability to warn against harmful choices. This allows users to interact with CareCart as they would with someone whom they trust, has a lot of accurate knowledge about products, and can provide immediate and personalized guidance. Another metaphor we implemented in our product is the concept of a cooking blog, where students and users can share tested products, recipes, and their experiences, replicating how communities naturally exchange food tips and budget hacks with others.

The second element is interactions, which consists of the different ways that users can engage with any system. Our interface includes many ways that allows users to interact with our product whilst reducing cognitive load. Our product offers a clean menu that makes it easy for students to navigate to where they want to go - whether that be to research a product, find relevant recipes, or check in with their community. We also offer multiple ways that users can find the products they are interested in via the search bar - they have the option of searching by products, or by store to check if their product of interest is beneficial for them, or to find an alternative grocery. Furthermore, users can engage through posting their own recipes, commenting and saving on alternative items and recipes, joining health-focused groups, and fostering a supportive community.

The last element is the interface, which consists of the format and layout of said interface. Our interface is designed for quick, on-the-go use for individual needs. Our dashboard interface allows for quick and easy access to personalized recommendations, a multi-use search bar, saved lists and recipes, access to community forums, all of which follow familiar media layout conventions. Lastly, as mentioned earlier, the search interface has many functions, allowing for users to rapidly discover a new product, find details about a certain product, find products in a certain store, all while using visual indicators for quick decision making.

Our conceptual model addresses the issue of overwhelming grocery shopping by transforming complicated nutritional decision-making into a product that allows for a simple, guided choice process. The personal assistant metaphor ensures that users are more comfortable and trusting while conducting health-conscious shopping, while community features offer social connection and practical solutions from other users facing similar challenges.

6 Prototyping

We approached our prototype step by considering the information we learned from interviewing our users. The information helped us

determine what features and design we wanted to focus on when creating the initial lo-fi prototype.

The figure above shows the start of the lo-fi prototype. We knew at the very beginning, it was important to create a welcome page that was able to introduce to the users what exactly the CareCart is meant for. Being able to have a welcome page with information on CareCart could give the user quick confirmation on if CareCart will be useful for them. We also knew we would need to incorporate a page that gave users the ability to log in or create an account. That way, they would be able to add their personal health concerns, grocery lists, recipes, and food items.

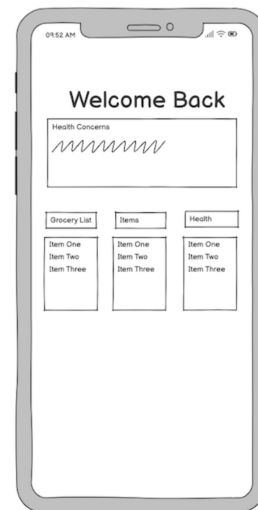


Figure 1: Lo-fi prototype: Main Dashboard.

The figure above shows the continuing process of our lo-fi prototype. We started to think about the next steps after a user creates an account or logs into an existing account. We then decided to create a main page after the user has logged in. The page would incorporate all of the main aspects of the user and basically, the main focuses of what CareCart is intended for. The page would include the user’s health concerns, their grocery lists, and saved food items.

The figure above shows the lo-fi of the different pages we decided would be important to implement. We wanted users to have the option to go further into being able to see recipes, get suggestions, or get more nutritional information on certain food items. We first created a page that would show the user’s grocery list. Based off of the list, we decided that recipe suggestions would be generated for the user. We included another page that focused on the aspect of recipes itself, giving the users a variety of recipes to choose from. We also decided to make a page that would focus on specific food items. This page would give nutrition information on specific food items and would give price information as well as the option to add these items to a list.

Finally, the figure above shows the last frame in our lo-fi prototype. We wanted the community page to be the overall “finale” of



Figure 2: Lo-fi prototype: Feature Pages (Grocery List, Recipes, Food Items).

CareCart. We decided this was best as users would have the information they need in order to share or ask questions about recipes and food overall. This frame shows our community page. Users would have the option to share recipes, talk about food, share healthy alternatives, and overall connect with others using CareCart.

Transitioning from the lofi sketches and midfi designs we worked on developing an interactive, hifi prototype. This prototype brought our design to life by not only a polished user interface and defined user flow but also with a fully functional and interactive product. The prototype begins with a landing page that clearly communicates CareCart’s mission and how it works. The design purposefully uses more grounded and natural colours leaning towards calm greens, muted oranges and warm brown tones. By using such a colour palette we brought focus more on the healthy and nourishing aspect of our mission.

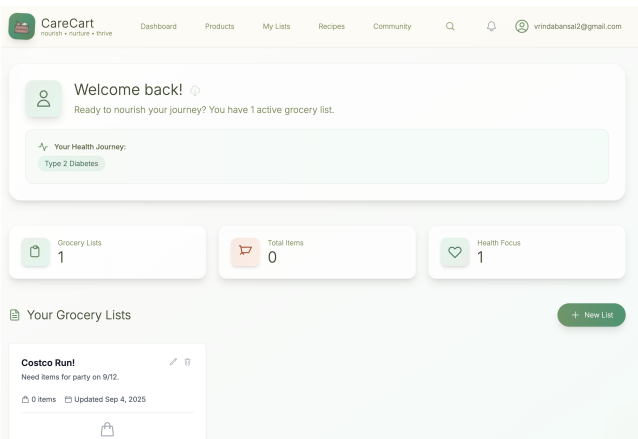


Figure 3: Hi-fi prototype: User Dashboard.

After the user logs in, they are brought to a personalized dashboard. This screen acts as a central hub, where the user can see a summary of their journey and possible action items. We kept in line with our original ideas by showing the user’s primary health focus and a summary of their current grocery lists. We hope that by providing such personalization the user feels like the platform is tailored to their needs from the second they log in.

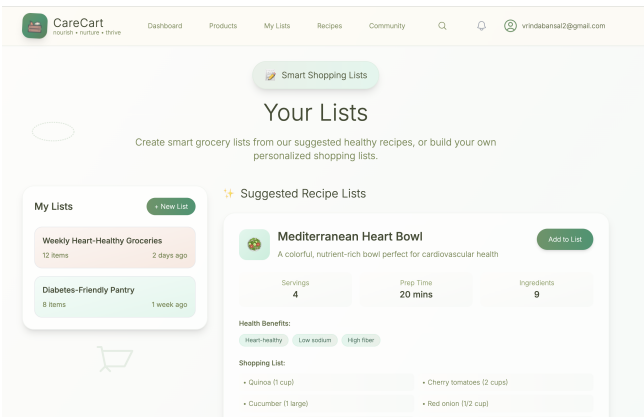


Figure 4: Hi-fi Prototype: Your Lists

From the dashboard the user is able to navigate to the other main pages such as Products, My Lists and Community using the nav bar at the top. The Products page is a core feature since it is the main page where users are able to find suitable foods. Here, users can successfully search for specific items or filter products from their favourite stores. Each product is given a product card that includes the nutritional information, company, store and healthy pros or cons. This feature implements our goal of providing personal nutritional guidance in a simple actionable format. Then, the “My Lists” section the user to view their saved lists and see possible suggestions. The integration of recipes and shopping lists here streamlines the entire process making it simple and easy for the user. Finally, the last part of the project is “Community” which brings our vision to life as it creates a supportive social network for different communities with different medical conditions. We had envisioned this earlier as the connect page and then improved it by adding suggested or popular communities to push users to check out this feature.

7 User testing

To evaluate the usability, emotional resonance, and function effectiveness of CareCart, we conducted structured user testing across both low-fidelity (lo-fi) and high-fidelity (hi-fi) prototypes. Our goal, was to determine whether our design met the needs of college students with dietary restrictions, health conditions, and financial constraints. To ensure diversity and reduce sample bias, we recruited six participants, deliberately avoiding close friends or prior collaborators. This allowed us to gather feedback from users with varied nutritional needs, including those dealing with PCOS, lactose intolerance, vegetarian diets, and general budget limitations. Throughout this process, we applied Norman’s Interaction Model to identify gaps in execution and evaluation, as well as Nielsen’s usability heuristics to guide iterative improvements.

During the low-fi testing phase, wireframes and storyboard walkthroughs were used to simulate core user flows. Using the Wizard-of-Oz method, a team member manually guided participants through the interface, allowing for observation of user behavior and collection of qualitative feedback without requiring a fully functional system. Participants completed three primary

tasks: setting up a health profile, searching for food items and receiving recommendations, and saving recipes while exploring the community page.

Findings from the lo-fi testing revealed several critical areas for improvement. Users expressed confusion about the welcome screen, which was overly verbose and did not communicate the purpose of the app effectively. Navigation across screens lacked clarity, and the sign-in process was also fragmented, requiring multiple steps that disrupted the user's flow. Participants also struggled with learnability, as page layouts did not intuitively guide them through the app's features. In the recipe section, the absence of filtering constraints made it difficult to locate relevant options, and the lack of personalized feedback based on health profiles diminished the perceived value of the recommendations. The community page, while conceptually appreciated, lacked user control, as participants wanted to join interest-based groups but were unable to do so.

In response, we refined our hi-fi prototype to address this. The updated version incorporated a streamlined welcome screen with a concise tagline and visual hierarchy that immediately conveyed the app's purpose, improved efficiency by integrating Google login, reducing friction in the onboarding process, we added short page descriptions and redesigned the navigation layout to follow a more intuitive structure. We also updated the recipe section with filters, as well as personalized feedback via health tags and dashboard cues, reinforcing the app's responsiveness to individual needs. Finally, the community page was redesigned to support tagging and group selection, enhancing user agency and emotional connection.

8 Discussion

From this project, we learned about the importance of human-centered design and its ability to solve everyday problems. When it comes to taking the initiative to solve everyday problems, it is important to put the main focus on the users themselves, especially the users that are actively experiencing the issue. In this project, our main focus was on students, especially those that experience health issues and a tight budget. In order to start finding the solution to our problem, it was important to approach the solution by gathering information from interviews, iterative testing, and with empathy. We also learned the importance of user testing and making sure we go above and beyond to create the best solution possible. The user testing must include a variety of users in order to remain as unbiased as possible.

9 Future Work

CareCart's current prototype lays a strong foundation for addressing student nutrition through human-centered design, with several extensions planned to enhance its impact. Future iterations could integrate real-time grocery data and barcode scanning to support in-the-moment decision-making and reduce cognitive load. Personalization may be expanded through adaptive features like symptom tracking, enabling the app to respond to users' evolving health conditions and routines. To encourage long-term engagement, gamified elements such as health badges, progress milestones, or recipe streaks could reward users for consistent, health-conscious choices. These features align with HCI principles of persuasive design and

intrinsic motivation. Prioritizing multilingual support would improve accessibility for students from diverse cultural and linguistic backgrounds. Together, these proposals reflect our continued commitment to designing with empathy, adaptability, and shared meaning—the core principles of human-computer interaction.

Acknowledgments

This section outlines the peer contributions for the project.

- **Priyanshi Singh:** 16.67 - Collaborated with group during topic selection & project idea; contributed significantly to the IPD; contributed heavily to the video presentation & script; contributed to the final report (conceptual model design mainly, but also contributed to the overall report content and formatting)
- **Vrinda Bansal:** 16.67 - Collaborated and provided base idea for CareCart; contributed significantly to the IPD; did 4 user interviews for feedback loops; Build fully functional prototype from scratch; contributed to final report (hifi content, report structuring in ACM format)
- **Ruoyu (Lindsay) Li:** 16.67 - Collaborate with group to brainstorm ideas for topic selection and frame the basic functions and meaning of Carcart; compose the first version of IPD and revise with teammates for finest version; carry out 5 TD steps like interviews for video presentation; contribute to final report with several parts (background, design method, and understanding the user)
- **Charishma Chandu:** 16.67 - Collaborated in refining the project topic; contributed to the IPD and its revision; designed and led user testing with 6 users for iterative feedback for lo-fi and hi-fi prototypes; contributed to the final report (User Testing and Future Work); supported integration of HCI principles; actively participated in video scriptwriting
- **Anusha Ramachandran:** 16.67 - Collaborated in refining project topic, contributed to the IPD and revision. Contributed to drafting the HCI method, process and motivation of CareCart. Led research for benefits of CareCart, our target audience, and challenges we faced. Supported integration of HCI principles and 5 steps of design thinking and actively participated in video scriptwriting
- **Andrea Zhang:** 16.67 - Collaborated with group to create a topic idea, worked on parts of writing up & developing the IPD, heavily contributed to creating the lo-fi wireframes, contributed to the prototype & discussion part of final report

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